

Diodes D2 and D3 are essentially the snubbers, and are used to protect the MOSFET from going above 600V.

Flyback transformer.

A flyback circuit is simply a pair of coupled inductors. If a current is passed through one inductor, it will store energy $E = \frac{1}{2} (L I^2)$, where 'L' stands for inductance in henry, and 'I' for current in amperes. This energy can later be taken out of the second inductor, which is coupled to the former at a different volt-current ratio. The flyback's energy storage and extraction mechanism is interesting. The key point is the polarity of the winding; the secondary is out of phase with the primary, as is evident in Fig. 2 (the dots indicate polarity).

When the MOSFET of IC1 is 'closed,' the current flows through L1. Point A on L1 goes positive and by transformer action, and considering the polarity of dots, point C on L2 goes negative. This reverse biases diode D4, and no current flows in the secondary winding. Similarly, when the MOSFET is 'open,' the current flow through L1 is interrupted and, by Lenz's Law, a voltage of polarity opposite to the applied voltage appears on L1 and L2. Thus, point A on L1 goes negative and point C on L2 goes positive. This situation forward biases diode D4. The energy stored in the core causes the current to flow through winding L2. This charges capacitor C2 and also powers the load. The charge on C2 is used in the next half of the cycle to keep the current through the load somewhat constant.

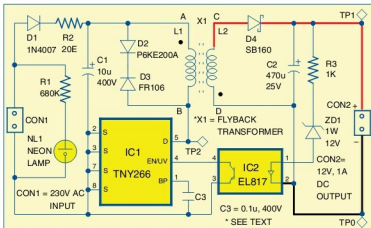


Fig. 2: Circuit of the 12V, 1A SMPS

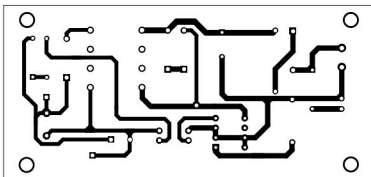


Fig. 3: An actual-size, single-side PCB for the 12V, 1A SMPS

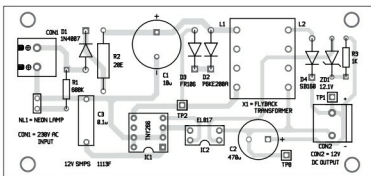


Fig. 4: Component layout for the PCB

TABLE II
Test Points

Test point	Details (without load)
TP0	0V, GND
TP1	12V
TP2	Around 120 kHz

The cycle repeats endlessly. The MOSFET is switched on/off continuously at a frequency of around 120 kHz to keep this process running.

The design data for the transformer

is as follows:

1. Duty cycle = 0.45 (max. duty cycle for DCM flyback = 0.5; less 10% safety margin)

2. Core saturation magnetisation $B_{sat} = 0.24T$

3. Core area of EE20 = $25mm^2$

Winding details computed for the SMPS are shown in Table I.

The feedback circuit.

Regulated output needs feedback to control the pulse width modulation (PWM) of the MOSFET. TNY266 has a fabulous control feature; it stops the switching cycle as soon as any current is taken out of pin 4 of the device. If SMPS output exceeds the zener breakdown voltage then ZD1 conducts. This lights the opto-LED and the opto-transistor grounds pin 4 of IC1, resulting in immediate stoppage of the switching cycle. Also, when the primary is conducting, diode D4 on the secondary side is reverse biased. At this time, if the voltage across D4 exceeds its reverse breakdown voltage, the SMPS will fail. Here we have used an SB160 Schottky diode

with breakdown voltage = 60V.

Connector CON2 provides 12V regulated DC supply.

Construction and testing

A general core-selection 'rule of thumb' for SMPS below 50 watts is 2-3 mm^2 core area per watt. For a primary input power of 16W, a core with a core area of 32-48 mm^2 is needed. EE20 core will work well for this design.

Transformer wire. Any wire that can carry the required current can be